

Lab 3

IMPORTANT INFO Large Language Model usage is ALLOWED for this assignment, but MUST be conducted through **Google Gemini** using your NCSU account. <https://oit.ncsu.edu/2024/11/20/gemini-chatbot-available-to-nc-state/>).

IF YOU USE GEMINI TO COMPLETE THIS LAB, you are required to submit your chat logs. Please follow the Lab3 Gemini Log Extraction guide for details on how to do so.

IF YOU USE GEMINI TO COMPLETE A QUESTION ON THIS LAB, please fill in the corresponding white square (☐) next to the question text.

Your submission should consist of this PDF with your answers filled in, and any additional pages of work you may need to complete each problem.

Question 1:

Part 1 You are a restaurant manager, and you are trying to decide how to set up the salad bar. You have 10 items, named i_0 through i_9 . The salad bar has the following shape, where O_{ij} represents an open slot for an item to be placed in:

$$\begin{array}{cccc} O_{00} & O_{01} & O_{02} & O_{03} \\ O_{10} & O_{11} & O_{12} & O_{13} \\ & O_{21} & O_{22} & \end{array}$$

1. How many different ways can you assign the items from the set $\{i_0, \dots, i_9\}$ to the open positions $O_{00}, \dots, O_{22}$, if every item must be present? ☐
2. ... if i_9 cannot be served due to allergies? (i.e only $\{i_0, \dots, i_8\}$ available) ☐
3. ... if i_8 and i_9 must be in the same row? ☐
4. ... in the same column? ☐

5. ... next to each other? $\square$

6. ... if the salad bar doesn't need to be full? $\square$

7. ... if $\square$

$\forall (i_x, i_y) \in i_0, \dots, i_9$ and their corresponding placements $i_x : O_{a,b}, i_y : O_{c,d}$

$$[x > y + 6] \rightarrow [a > c]$$

8. ... if only 5 i_1 are ready to be served and nothing else is ready? (items are now the multi-set $\{i_1, i_1, i_1, i_1, i_1\}$) $\square$

9. ... if only 5 i_1 and 5 i_2 are ready to be served and nothing else is ready? $\square$

10. ... if only 3 i_1 and 3 i_2 3 i_3 and 1 i_4 are ready to be served and nothing else is ready? $\square$

11. ... if each possible placement O_{ij} can now accommodate two items, without any notion of an ordering (i.e indistinguishable) $\square$
12. ... if items i_0, i_1, i_2 are warm and need to be placed in a warm spot on the top row? (placement $O_{0j} \forall j$) $\square$

Question 2

Consider the following axioms describing the construction and properties of the natural numbers. Any solution that cites any property of natural numbers not included in axioms 1-6 will be considered incorrect.

- **Axiom 1** $0 \in \mathbb{N}$
- **Axiom 2** The successor for a natural number is defined as $S(n) = n + 1$
- **Axiom 3** If $n \in \mathbb{N}$, then $S(n) \in \mathbb{N}$
- **Axiom 4** If the successors of two natural numbers are equal, then the numbers themselves are equal. $\forall m, n \in \mathbb{N} [S(n) = S(m)] \rightarrow [n = m]$
- **Axiom 5** 0 is not the successor of any natural number. $\forall n \in \mathbb{N}, S(n) \neq 0$
- **Axiom 6** For any $m, n \in \mathbb{N}$: $m + S(n) = S(m + n)$

Part 1 Prove by induction that for any natural number $n \in \mathbb{N}$, $n + 0 = n$ $\square$

Part 2 Prove by induction that for all natural numbers $m, n \in \mathbb{N}$, their sum $m + n \in \mathbb{N}$. $\square$

(**HINT** Let m be a fixed but arbitrary number (i.e universal instantiation), and then induct on n . Then apply universal generalization to recover the intended conclusion.)

Part 3 Now consider the following construction and properties of the **Squemp** Numbers, defined by the following axioms and denoted by $\mathbb{S}$:

- **Axiom 1** $0 \in \mathbb{S}$
- **Axiom 2** The successor for a **squemp** number is defined as $S(n) = 2n + 3$
- **Axiom 3** If $n \in \mathbb{S}$, then $S(n) \in \mathbb{S}$
- **Axiom 4** If the successors of two squemp numbers are equal, then the numbers themselves are equal. $\forall m, n \in \mathbb{S} [S(n) = S(m)] \rightarrow [n = m]$
- **Axiom 5** 0 is not the successor of any squemp number. $\forall n \in \mathbb{S}, S(n) \neq 0$

Use formal proof techniques to prove or disprove the validity of the following property of the **Squemp** numbers: $\square$

$$\forall m, n \in \mathbb{S} : m + S(n) = S(m + n)$$

Part 5 Are the **Squemp** numbers closed under addition? $\square$

$m, n \in \mathbb{S}$, their sum $m + n \in \mathbb{S}$

Logical Implications

1. $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ Hypothetical Syllogism
2. $[(p \vee q) \wedge \neg p] \rightarrow q$ Disjunctive Syllogism
3. $[p \wedge (p \rightarrow q)] \rightarrow q$ Modus Ponens
4. $[(p \rightarrow q) \wedge \neg q] \rightarrow \neg p$ Modus Tollens
5. $(p \rightarrow 0) \rightarrow \neg p$ Absurdity
6. $(p \wedge q) \rightarrow p$ Simplification
7. $[(p \rightarrow q) \wedge (r \rightarrow s)] \rightarrow [(p \vee r) \rightarrow (q \vee s)]$ Constructive Dilemma ($\vee$)
8. $[(p \rightarrow q) \wedge (r \rightarrow s)] \rightarrow [(p \wedge r) \rightarrow (q \wedge s)]$ Constructive Dilemma ($\wedge$)
9. $p \rightarrow (p \vee q)$ Addition
10. $[p \wedge q] \rightarrow (p \wedge q)$ Conjunction

Logical Equivalences

11. $\neg\neg p \equiv p$ Double Negation
12. $(p \vee q) \equiv (q \vee p)$ Commutativity ($\vee$)
13. $(p \wedge q) \equiv (q \wedge p)$ Commutativity ($\wedge$)
14. $[(p \vee q) \vee r] \equiv [p \vee (q \vee r)]$ Associativity ($\vee$)
15. $[(p \wedge q) \wedge r] \equiv [p \wedge (q \wedge r)]$ Associativity ($\wedge$)
16. $[p \vee (q \wedge r)] \equiv [(p \vee q) \wedge (p \vee r)]$ Distributivity ($\vee$ over $\wedge$)
17. $[p \wedge (q \vee r)] \equiv [(p \wedge q) \vee (p \wedge r)]$ Distributivity ($\wedge$ over $\vee$)
18. $(p \vee p) \equiv p$ Idempotence ($\vee$)
19. $(p \wedge p) \equiv p$ Idempotence ($\wedge$)
20. $(p \vee 0) \equiv p$ Identity ($\vee$)
21. $(p \wedge 1) \equiv p$ Identity ($\wedge$)
22. $(p \wedge 0) \equiv 0$ Domination ($\wedge$)
23. $(p \vee 1) \equiv 1$ Domination ($\vee$)
24. $(p \rightarrow p) \equiv 1$ Domination ($\rightarrow$)
25. $(p \vee \neg p) \equiv 1$ Complement (Tautology)
26. $(p \wedge \neg p) \equiv 0$ Complement (Contradiction)
27. $\neg(p \vee q) \equiv (\neg p \wedge \neg q)$ DeMorgan's ($\vee$)
28. $\neg(p \wedge q) \equiv (\neg p \vee \neg q)$ DeMorgan's ($\wedge$)
29. $(p \rightarrow q) \equiv (\neg q \rightarrow \neg p)$ Contrapositive
30. $(p \rightarrow q) \equiv (\neg p \vee q)$ Conditional Identity ($\rightarrow$)
31. $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$ Conditional Identity ($\leftrightarrow$)
32. $[(p \rightarrow r) \wedge (q \rightarrow r)] \equiv [(p \vee q) \rightarrow r]$ Implication
33. $[(p \rightarrow q) \wedge (p \rightarrow r)] \equiv [p \rightarrow (q \wedge r)]$ Implication